

Autodesk® Scaleform®

Scaleform 3D

Scaleform 3.2

3D

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Autodesk® Scaleform® 4.3

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Scaleform 3D

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1

Scaleform® 3Di™

UI

3D Flash

Scaleform 3.2 ,Scaleform

ActionScript ,

3D

Scaleform 3.2 ActionScript 2.0 ,

3D AS --- AS 3.0

HUD (_z _zscale _xrotation _yrotation _matrix3d _perspfov),
UI 3D 3D

c + +

3D

Scaleform

API

API

3D Gfx::Value

, 3D 3D Flash ,

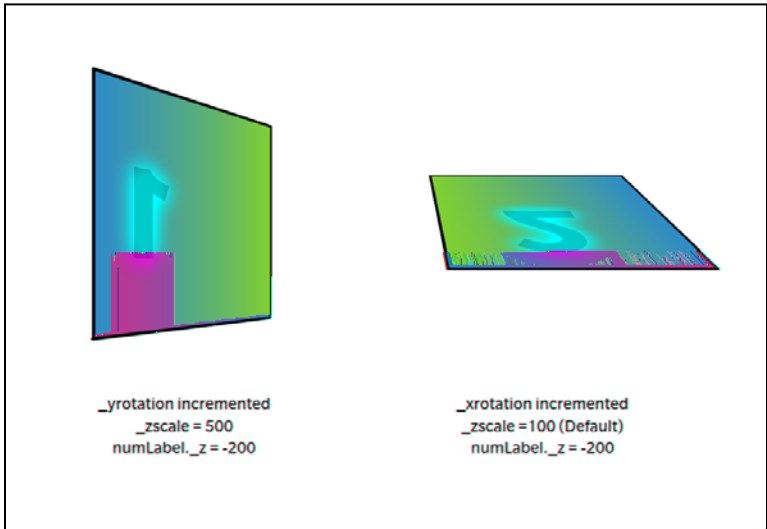
2 Flash 10

Scaleform 3Di Flash 10 3D , 3Di

2.1

Flash 10 3D , 3Di :

- 1. , , ,
- 2. (Back-face culling)



1 (Culled)

2.2

Scaleform 3Di , Flash 10 3D ,

2D 3D , 3D 3D
Flash 10

3 3D Flash

Scaleform 3.2, 3D Flash

3.1

Flash , 3D Flash
Scaleform 3.2 , 3D Flash
“(Render to Texture) 3Di Flash
3D , 3D , 3D
Z() / (UI)
,
Flash (Scaleform) ,
, Flash ,
, “SWF “(SWF to Texture) , Scaleform SDK

3.2 3Di

4 3Di API

4.1 ActionScript 2 API

3Di,Flash

```

    ,
    ,
    3D , 3D (x y z
    3d ),
    3D ,
    3d " "(Null)( , ActionScript ,foo._matrix3d = null),
    2D z 0 3D
    ,3D 3 (x y z ) , Z 2D
    ,3D , _zscale ActionScript

```

Scaleform 3D Flash 10 +Z () +Z ,FOV 55 , +X ,+Y ,

3D Flash

-
-
-

3Di ActionScript

- **_z** - Z (), 0
 - **_zscale** - Z , 100
 - **_xrotation** - X() , 0
 - **_yrotation** - Y() , 0
 - **_matrix3d** - 16 (4 x 4) 3D
NULL, 3D , 2D
 - **perspfov** - (FOV) > 0 < 180
-1 FOV
- 55

ActionScript

gfxExtensions

True()

,
_xrotation _yrotation 3D
3Di

1

Y 45

```
_global.gfxExtensions = true;  
mc1._yrotation = 45;
```

2

X Z

```
_global.gfxExtensions = true;  
sq1._zscale = 500;
```

```
function rotateAboutXAxis(mc:MovieClip):Void  
{  
    mc._xrotation = mc._xrotation + 1;  
    if (mc._xrotation > 360)  
    {  
        mc._xrotation = 0;  
    }  
}
```

```
onEnterFrame = function()  
{  
    rotateAboutXAxis(sq1);  
}
```

3

3D 3D

```
function PixelsToTwips(iPixels):Number
```

```

{
    return iPixels*20;
}

function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [ 1, 0, 0, 0,
                          0, 1, 0, 0,
                          0, 0, 1, 0,
                          tX,tY,tZ,1];
    return matrixC;
}

this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);

```

4.2 C++ 3Di

4.2.1 Matrix4x4

Matrix4x4 4x4 3D
() , Render_Matrix4x4.h

4.2.2 Direct Access

Direct Access (Gfx::Value) 3Di API 3D
3Di ActionScript

, Gfx_Player.h [Gfx::Value::DisplayInfo](#)

```
void SetZ(Double z)
void SetXRotation(Double degrees)
void SetYRotation(Double degrees)
void SetZScale(Double zscale)
void SetFOV(Double fov)
void SetProjectionMatrix3D(const Matrix4F *pmat)
void SetViewMatrix3D(const Matrix3F *pmat)
bool SetMatrix3D(const Render::Matrix3F& mat)
```

```
Double GetZ() const
Double GetXRotation() const
Double GetYRotation() const
Double GetZScale() const
Double GetFOV() const
const Matrix4F* GetProjectionMatrix3D() const
const Matrix3F* GetViewMatrix3D() const
bool GetMatrix3D(Render::Matrix3F* pmat) const
```

4.2.3 Render::TreeNode

Render::TreeNode , 3D " "(View) "
"(Perspective) , ,

Gfx_Player.h

```
void    SetProjectionMatrix3D(const Matrix4F& m)
void    SetViewMatrix3D(const Matrix3F& m);
```

4.2.4 : Direct Access API

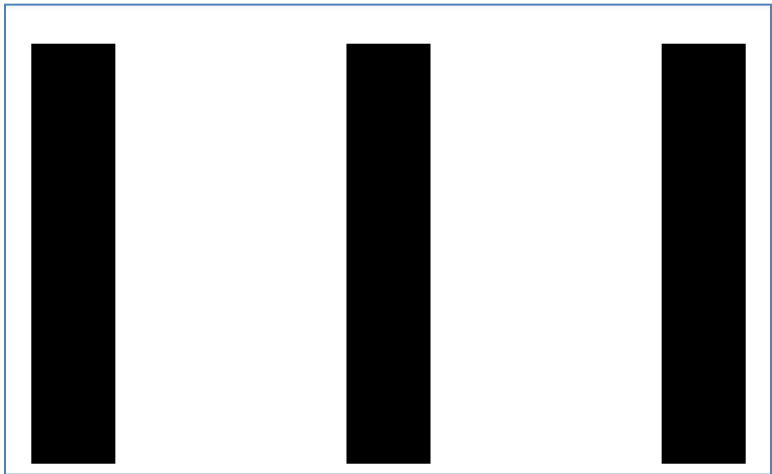
```
Ptr<GFx::Movie> pMovie = ...;
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
    GFx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        // set perspectiveFOV to 150 degrees
        Double oldPersp = dinfo.GetFOV();
        dinfo.SetFOV(150);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
```

5

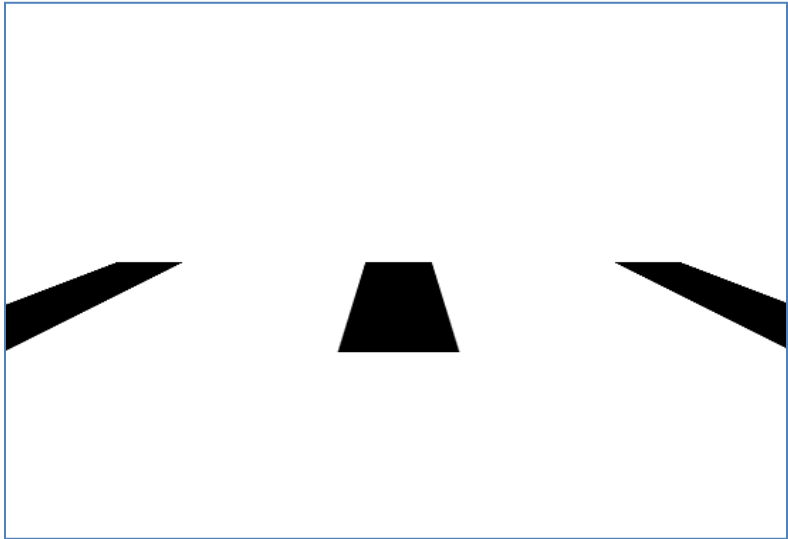
Direct Access API (FOV)

FOV 0, FOV

z 0 180, 1, FOV 179 ,



2 Flash



3 -80 55 X

```

        FOV (5 )
        .
        ,
        ○   ActionScript      property _perspfov      -1      C++
        Direct Access API      :

// 1.   Perspective FOV      -1

Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
    Gfx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        dinfo.SetPerspFOV(-1);
        tmpVal.SetDisplayInfo(dinfo);
    }
}

//      2.

void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)
{
    const float nearZ = 1;
    const float farZ = 100000;
    float DisplayHeight = fabsf(visFrameRectInTwips.Height());
    float DisplayWidth = fabsf(visFrameRectInTwips.Width());
    matPersp->OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root.mc");
if (bOK)
{
    Gfx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        Matrix4F perspMat;
        MakeOrthogProj(PixelsToTwips (pMovie->GetVisibleFrameRect()), &perspMat);
        dinfo.SetProjectionMatrix3D(&perspMat);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
}

```

(Projection)

```
Ptr<Movie> pMovie = ...
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie->GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()), fabs(visFrameRectInTwips.Height()),
              nearZ, farZ);
pMovie->SetProjectionMatrix3D(ortho);
```

5.1 AS3

ActionScript 3 , PerspectiveProjection()

Example:

```
import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;
par.y = 100;
par.rotationX = 20;

addChild(par);

var chRed:Sprite = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0, 0, 100, 100);
chRed.x = 50;
chRed.y = 50;
chRed.z = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection = pp3;
```

6 Stereoscopic 3D

Scaleform 3Di	UI,	3D
---------------	-----	----

```
Scaleform 3Di          3D          ,          ,PC          NVIDIA 3D Vision Kit  NVIDIA
          ,          ,          ,          (          )
          ,          Display          ,          ,
          (Parallax Shift)  Scaleform 4          ,          PS3
TinyPlayerStereo(      Scaleform          ):
Apps\Samples\GFxPlayerTiny\GFxPlayerTinyPS3GcmStereo.cpp
```

6.1 NVIDIA 3D Vision

6.1.1 3D Vision

	NVIDIA 3D Vision Kit	3Di UI, 3D	NVIDIA (Stereoscopic 3D
Test)			

6.1.2

Scaleform Player 3D Vision

Direct3D Scaleform Player, Scaleform Player

command line argument -f NVIDIA

3D 3D (CTRL + T)

6.1.3

3D Vision (Convergence)

3D Vision (CTRL + F3) / (CTRL + F4) IR

Scaleform Player (CTRL + F5) (CTRL + F6),

Shortcuts" NVIDIA "Stereoscopic 3D", "Set up stereoscopic 3D", "Set Keyboard (CTRL + F5) (CTRL + F6) "(Enable the advanced in-game settings) (CTRL + F7)

F5), (CTRL + F5) (CTRL + F6), 10 15 (/) (CTRL + 0 --- (CTRL + F7)

NVIDIA API, 3D

API
<http://developer.nvidia.com/object/nvapi.html>.

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform Stereo API

Scaleform 4.1 Render::HAL API, 3D 3D
HDMI 1.4 .

, Scaleform , 6.2.1 .

Scaleform 4

Render::Viewport::SetStereoViewport

SF4 (Viewport Stereo)

```
//    ;    ,  
  
View_Stereo_SplitV    = 0x40,  
View_Stereo_SplitH    = 0x80,  
View_Stereo_AnySplit    = 0xc0,  
  
void SetStereoViewport(unsigned display);
```

6.2.1 Scaleform Stereo

```
Render::StereoParams s3DInfo;  
s3DInfo.DisplayDiagInches = 46;    // 46 inch TV  
s3DInfo.DisplayAspectRatio = 16.f / 9.f;  
s3DInfo.Distortion = .75;    // 0 to 1  
s3DInfo.EyeSeparationCm = 6.4;    // this will default 6.4cm  
pRenderHAL->SetStereoParams(s3DInfo);
```

, Display , , Display
Scaleform :

```
pMovie->Advance(delta);  
  
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);  
pMovie->Display();
```

```
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);  
pMovie->Display();
```

7

3Di

Scaleform SDK

3D Flash

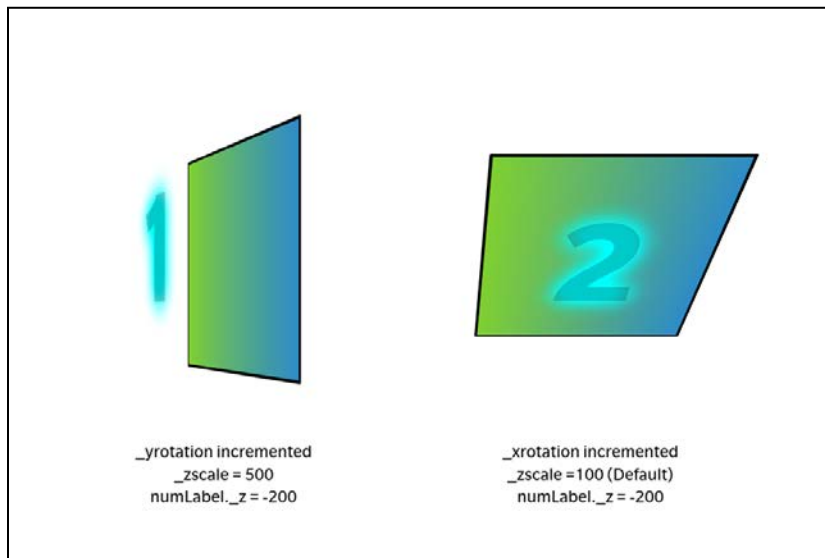
(SWF FLA)

,
AS3\Samples

C:\Program Files (x86)\Scaleform\GFx SDK 4.3\Bin\Data\AS2 or



4 3DGenerator



5 3D



6 3D